

JDScience BTEC Level 3 Waves in Communication Answer Sheet

Unit 1C - wave properties, wave equation, phase, interference and diffraction.

Mark scheme

Task	Answer / marking points
A	Correct definitions: oscillation = repeated vibration; compression = particles close together; rarefaction = particles spread apart; amplitude = maximum displacement; frequency = waves per second; period = time for one wave; wavelength = distance between matching points.
B	1) 20,000 m/s. 2) 2000 m/s. 3) 20 Hz. 4) 2 m.
C	In the same medium, sound waves travel at approximately the same speed; higher pitch means higher frequency and shorter wavelength, not faster arrival.
D	Constructive: whole-number wavelength path difference, waves arrive in phase and amplitude increases. Destructive: half-number wavelength path difference, waves arrive out of phase and cancel/reduce.
E	Central maximum is bright because waves have zero path difference and interfere constructively. Other maxima occur where path difference is $n\lambda$; minima occur where waves arrive out of phase.